Day-10 absolute zonal power spectra | 11 fields, physical units, log–log axes
8 matched 2023 starts; 30–60° latitude in both hemispheres; late SWA

T2m — 2 m temperature
Power (K²)
Zonal wavenumber k

U10 — 10 m zonal wind
Power ((m/s)²)
Zonal wavenumber k

V10 — 10 m meridional wind
Power ((m/s)²)
Zonal wavenumber k

MSLP — Sea-level pressure
Power (hPa²)
Zonal wavenumber k

TP6 — 6 h precipitation
Power (mm²)
Zonal wavenumber k

T850 — 850 hPa temperature
Power (K²)
Zonal wavenumber k

Q700 — 700 hPa specific humidity
Power ((g/kg)²)
Zonal wavenumber k

U850 — 850 hPa zonal wind
Power ((m/s)²)
Zonal wavenumber k

V850 — 850 hPa meridional wind
Power ((m/s)²)
Zonal wavenumber k

WS10 — 10 m wind speed
Power ((m/s)²)
Zonal wavenumber k

WS850 — 850 hPa wind speed
Power ((m/s)²)
Zonal wavenumber k

GC
MSE continuation
MGDA
ERA5

Day-10 spectral power relative to ERA5 | closer to 1 is better
8 matched 2023 starts; 30–60° latitude in both hemispheres; late SWA

T2m — 2 m temperature

U10 — 10 m zonal wind

V10 — 10 m meridional wind

MSLP — Sea-level pressure

TP6 — 6 h precipitation

T850 — 850 hPa temperature

Q700 — 700 hPa specific humidity

U850 — 850 hPa zonal wind

V850 — 850 hPa meridional wind

WS10 — 10 m wind speed

WS850 — 850 hPa wind speed

Forecast power / ERA5 power

Zonal wavenumber k

GC

MSE continuation

MGDA

ERA5 target

Power-spectrum error across the rollout | lower is better
8 matched 2023 starts; 30–60° latitude in both hemispheres; late SWA

T2m — 2 m temperature

U10 — 10 m zonal wind

V10 — 10 m meridional wind

MSLP — Sea-level pressure

TP6 — 6 h precipitation

T850 — 850 hPa temperature

Q700 — 700 hPa specific humidity

U850 — 850 hPa zonal wind

V850 — 850 hPa meridional wind

WS10 — 10 m wind speed

WS850 — 850 hPa wind speed

Mean relative power error (%)

Forecast lead (days)

GC

MSE continuation

MGDA